

ZIHAN DONG

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EDUCATION

Rutgers University

Ph.D. in Statistics

Piscataway, NJ

2023–present

School of the Gifted Young,

University of Science and Technology of China (USTC)

B.S. in Mathematics, Major GPA: 4.17/4.30

Hefei, China

2019–2023

PREPRINTS/PUBLICATIONS

- Conference Papers

1. **Zihan Dong**, Zhixian Zhang, Yang Zhou, Can Jin, Ruijia Wu, Linjun Zhang (2026). Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Mathematical Reasoning via Comparative Signals. *ICML 2026*.
2. Tianci Liu, **Zihan Dong**, Linjun Zhang, Haoyu Wang, Jing Gao, Emre Kiciman, Ranveer Chandra, and Wei-Ting Chen (2026). Personalization as Inverse Planning: Learning Latent Design Intents for Agentic Slide Generation via Structural Denoising. *ECCV 2026*.
3. Ran Xu, Yuchen Zhuang, **Zihan Dong**, Ruiyu Wang, Yue Yu, Joyce C. Ho, Linjun Zhang, Haoyu Wang, Wenqi Shi, and Carl Yang (2025). AceSearcher: Bootstrapping Reasoning and Search for LLMs via Reinforced Self-Play. *NeurIPS 2025 Spotlight*.
4. Ran Xu, Tianci Liu, **Zihan Dong**, Tony Yu, Ilgee Hong, Carl Yang, Linjun Zhang, Tuo Zhao, Haoyu Wang (2026). Alternating Reinforcement Learning for Rubric-Based Reward Modeling in Non-Verifiable LLM Post-Training. *ICML 2026*.
5. Shuhang Lin, Chuhao Zhou, Xiao Lin, **Zihan Dong**, Kuan Lu, Zhencan Peng, Jie Yin, Dimitris N. Metaxas (2026). Conformal Path Reasoning: Trustworthy Knowledge Graph Question Answering via Path-Level Calibration. *ICML 2026*.
6. Peng Xia, Jinglu Wang, Yibo Peng, Kaide Zeng, **Zihan Dong**, Xian Wu, Xiangru Tang, Hongtu Zhu, Yun Li, Linjun Zhang, Shujie Liu, Yan Lu, and Huaxiu Yao. (2026). MMedAgent-RL: Optimizing Multi-Agent Collaboration for Multimodal Medical Reasoning. *ICLR 2026*.
7. Tianci Liu, Ruirui Li, **Zihan Dong**, Hui Liu, Xianfeng Tang, Qingyu Yin, Linjun Zhang, Haoyu Wang, and Jing Gao (2025). Mitigating Heterogeneous Token Overfitting in LLM Knowledge Editing. *ICML 2025*.
8. Difei Gu, Yunhe Gao, Gerasimos Chatzoudis, **Zihan Dong**, Guoning Zhang, Bangwei Guo, Yang Zhou, Mu Zhou, Dimitris N. Metaxas (2026). LUCID-SAE: Learning Unified Vision-Language Sparse Codes for Interpretable Concept Discovery.
9. Pei Tian, **Zihan Dong**, Tianci Liu, Linjun Zhang, and Haoyu Wang (2026). RIMS: Preference Optimization via Smoothed Multi-pair Aggregation for Small-Scale LLM Retrieval-Augmented Generation. Submitted to *COLM 2026*.
10. Jiaqi Liu, Shi Qiu, Mairui Li, Bingzhou Li, Haonian Ji, Siwei Han, Xinyu Ye, Peng Xia, **Zihan Dong**, Meng Chen, Congyu Zhang, Letian Zhang, Guiming Chen, Haoqin Tu, Xinyu Yang, Lu Feng, Xujiang Zhao, Haifeng Chen, Jiawei Zhou, Xiao Wang, Weitong Zhang, Hongtu Zhu, Yun Li, Jieru Mei, Hongliang Fei, Jiaheng Zhang, Linjie Li, Linjun Zhang, Yuyin Zhou, Sheng Wang, Caiming Xiong, James Zou, Zeyu Zheng, Cihang Xie, Mingyu Ding, and Huaxiu Yao (2026). **AutoResearchClaw: Self-Reinforcing Autonomous Research with Human-AI Collaboration**. Submitted to *NeurIPS 2026*. <https://arxiv.org/abs/2605.20025>.
11. Yang Zhou, Can Jin, **Zihan Dong**, Zhepeng Wang, Yanting Yang, Shiyu Zhao, Lei Li, Runxue Bao, Yaochen Xie, and Dimitris N. Metaxas (2026). DARE: Difficulty-Adaptive Reinforcement Learning with Co-Evolved Difficulty Estimation. Submitted to *NeurIPS 2026*. <https://arxiv.org/abs/2605.09188>.

12. Yang Zhou, **Zihan Dong**, Zhenting Wang, Can Jin, Shiyu Zhao, Bangwei Guo, Difei Gu, Linjun Zhang, Mu Zhou, and Dimitris N. Metaxas (2026). Evidence Over Plans: Online Trajectory Verification for Skill Distillation. Submitted to *NeurIPS 2026*. <https://arxiv.org/abs/2605.09192>.
13. Haoxiang Jiang, **Zihan Dong**, Tianci Liu, Wanying Wang, Ran Xu, Tony Yu, Linjun Zhang, and Haoyu Wang (2026). RUBRIC-ARROW: Alternating Pointwise Rubric Reward Modeling for LLM Post-training in Non-verifiable Domains. Submitted to *EMNLP 2026*. <https://arxiv.org/abs/2605.29156>.
14. Can Jin, Yang Zhou, Qixin Zhang, Hongwu Peng, Di Zhang, **Zihan Dong**, Marco Pavone, Ligong Han, Zhang-Wei Hong, Tong Che, Dimitris N. Metaxas (2026). AIRL-S: Unifying Reinforcement Learning and Search-Based Test-Time Scaling via Adversarial Inverse Reinforcement Learning. Submitted to *EMNLP 2026*.

- Journal Papers

1. **Zihan Dong**, Ruijia Wu, and Linjun Zhang (2026). Labels or Preferences? Budget-Constrained Learning with Human Judgments over AI-Generated Outputs. Submitted to *Journal of the Royal Statistical Society, Series B (Statistical Methodology)*. <https://arxiv.org/pdf/2601.13458>.
2. **Zihan Dong**, Xin Zhou, Ryumei Nakada, Lexin Li, and Linjun Zhang (2025). Contrastive Network Representation Learning. Submitted to *JASA Theory and Methods*. <https://arxiv.org/pdf/2509.11316>.

INDUSTRY EXPERIENCE

Research Intern

May 2026 – Present

Qwen, Tongyi Lab, Alibaba Group

- Contributed to Qwen LLM post-training, including Supervised Fine-Tuning (SFT) data synthesis and reinforcement learning (RL) training for agentic coding and reasoning tasks.

PROJECTS

- **AutoResearchClaw**

13.5K+ GitHub stars

Self-Reinforcing Autonomous Research with Human-AI Collaboration.

TECHNICAL SKILLS

- Proficient with verl, SGLang, vLLM, and Megatron-LM for large-scale LLM post-training, distributed training, and high-throughput inference.
- 4+ Years of research experience in data science, statistics, and machine learning, with expertise in Python (PyTorch), R and SQL. Experienced in generative and foundation models, applying supervised and unsupervised learning methods in AI.
- Programming Languages: Python, R, C/C++, SQL
- Libraries and Tools: PyTorch, Scikit-Learn, Pandas, Numpy, Scipy, Matplotlib, Excel

HONORS AND AWARDS

- Best Ph.D. Qualifying Exam Performance Award, Rutgers University 2024
- National Scholarship, Top 1% nationwide, Chinese Ministry of Education 2022
- Cyrus Tang Scholarship, Top 3%, Cyrus Tang Foundation, Nationwide 2019-2022
- The 12th Chinese Mathematics Competitions, Second Prize in Professional Group 2020